BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Primary Storage: RAM, ROM and Cache

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. After giving a technical point in an explain answer, what should follow?

2. A = 1 and B = 0. What is A AND B?

A) 1 B) 2
C) 10 D) 0

3. A truth table has independent inputs A, B and C. Give the number of input combinations and list them in ascending binary order.

4. A loop should run while finished is false. It uses finished as its condition. Correct the condition. Explain what happens when finished=false and finished=true.

Today's objectives

Describe the roles and volatility of RAM, ROM and cache. Explain how these types of memory support a running computer.

Key vocabulary - Read, cover, recall

RAM
Volatile primary storage used for active programs and data.

ROM
Non-volatile memory typically storing firmware.

Cache
Small, fast memory close to the CPU.

Volatile
Loses stored contents when power is removed.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

RAM is working memory	RAM stores programs and data currently in use. It is volatile: its contents are lost when power is removed.
ROM and startup	ROM is non-volatile and typically stores essential firmware or startup instructions.
Why cache exists	Cache is small, very fast memory close to the processor, holding frequently or recently used data and instructions.
Compare the roles	RAM is general working memory; ROM retains firmware; cache reduces repeated slower memory accesses.

I Do - One worked example

A laptop uses RAM, ROM and cache. Explain the role of each and why cache helps the CPU. [6 marks]

1. Identify the memory purpose

2. RAM stores active programs/data; ROM retains firmware across shutdown.

3. Cache holds frequently used instructions/data close to CPU, reducing RAM access.

4. RAM and cache are volatile; ROM retains contents without power. [2 marks for each correctly explained role]

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Explain why opening more applications increases demand for RAM. [4]

2. A device must retain startup code when unplugged. Explain why ROM is suitable. [4]

3. Explain why CPU cache is usually smaller than RAM but helps performance. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the comparison so the application only loads if it has enough free RAM.

```
freeMB = 800
neededMB = 800
if freeMB > neededMB then
print("Load")
else
print("Wait")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State whether RAM is volatile and describe what happens to its contents when power is lost. [2]

Q3. Explain why a CPU benefits from cache memory. [3]

Q4. Compare RAM and ROM using purpose and volatility. [4]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. An algorithm only gives reliable results when its inputs are handled as specified. ☐ T ☐ F
2. RAM stores programs and data currently in use ☐ T ☐ F
3. ROM is non-volatile and typically stores essential firmware or startup instructions. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A linked reason or effect that answers the stated situation. ☐

Q2 D - 0 ☐

Q3 8: 000, 001, 010, 011, 100, 101, 110, 111. ☐

Q4 Use NOT finished. It is true when finished is false, so the loop continues. It is false when finished is true, so the loop stops. The original does the reverse. ☐

Code First - Model

```
freeMB = 800
neededMB = 800
if freeMB >= neededMB then
print("Load")
else
print("Wait")
endif
```

Output: Load

We Do - Check the model

1. Each open program requires instructions/data in RAM [1]. More active applications occupy more RAM [1]. Available RAM decreases [1], potentially requiring virtual memory [1]. ☐

2. ROM retains its contents without power [1], so firmware remains available after shutdown [1]. Startup instructions can be accessed when switched on [1], allowing the device to initialise [1]. ☐

3. Cache provides faster access than RAM [1] for frequently needed data [1], reducing time waiting for memory [1]. It is more expensive per unit of storage, so is generally smaller [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Volatile [1]; contents are lost when power is removed [1]. ☐

Q3 Frequently used data/instructions stored in cache [1]; CPU can access them faster than RAM [1], improving speed/reducing waits [1]. ☐

Q4 RAM stores currently running programs/data [1] and is volatile [1]; ROM typically stores firmware/startup instructions [1] and is non-volatile [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: